

A photograph of a hospital room with a patient bed, medical equipment, and a window. A semi-transparent teal overlay covers the center of the image, containing the title text.

Immigration Enforcement on Health Care Utilization

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Intro and Research Question

**Methodology, Data Sources, and
Model Description**

Exploratory Data Analysis

Results and Conclusion

Q&A

Introduction

Research Question: Does increased ICE enforcement have a causal effect with reduced health care utilization?

Why it matters:

In January 2025, policy changes reversed protections designating hospitals as "sensitive locations," potentially increasing fears in communities with intensified ICE activity.

Immigration-related fears negatively affect health care access in the U.S. (Kaiser Family Foundation 2025). Nearly 48% of likely undocumented immigrant adults and 14% of immigrant adults overall report avoiding medical care due to immigration concerns.

Hypothesis: Higher levels of local enforcement activity are associated with greater health care avoidance.

Methodology

We performed the following:



Data Collection

- CDC PLACES
- Transactional Records Access Clearinghouse (TRAC)
- American Community Survey (ACS)



Data Cleaning and EDA

- Merged data sources into ZCTA-year panel (2018-2023)
- Explored distributions and correlations of key variables
- Removed incomplete observations



Modeling

- Used one two-way fixed effects model on binary presence
- Assessed three One-Way Fixed Effect spatial models: log distance, regional radius, and distance decay.
- Included VIF-tested socioeconomic controls and clustered standard errors.



Analysis

- Descriptive and Spatial Analysis of variables
- Models evaluated based on R^2 and within R^2
- Post-Hoc Population Confounder Evaluation

Data Sources (ZCTA-level, 2018-2023)

CDC PLACES

Did you have a check-up in the past year?

- Model-based estimates of health outcomes, behaviors, and social needs from the CDC to support public health initiatives

Transactional Records Access Clearinghouse (TRAC)

How many people are detained by ICE in the area?

- Average daily population of ICE detention facilities

American Community Survey (ACS) Data

What are the social determinants of health?

- Race, citizenship status, income, education, employment status, and housing condition

Statistical Model

$$Y_{it} = \beta_0 + \beta_1(\text{Enforcement})_{it} + \Gamma \mathbf{X}_{it} + \alpha_i + \delta_t + \epsilon_{it}$$

Two-Way Fixed Effects (TWFE) Modeling

- Y_{it} : Outcome variable — percentage of adults who had a routine checkup in ZCTA i during year t .
- **Enforcement_{it}** : Primary independent variable (e.g., proximity to an ICE facility or detention population density).
- $\Gamma \mathbf{X}_{it}$: Vector of time-varying socioeconomic controls (race, citizenship, poverty, etc.).
- α_i : ZCTA-level fixed effects, capturing all time-invariant geographic heterogeneity.
- δ_t : Year fixed effects, capturing national macro-level shocks.
- ϵ_{it} : Idiosyncratic error term, with standard errors clustered at the ZCTA level.

Exploratory Data Analysis

Correlation

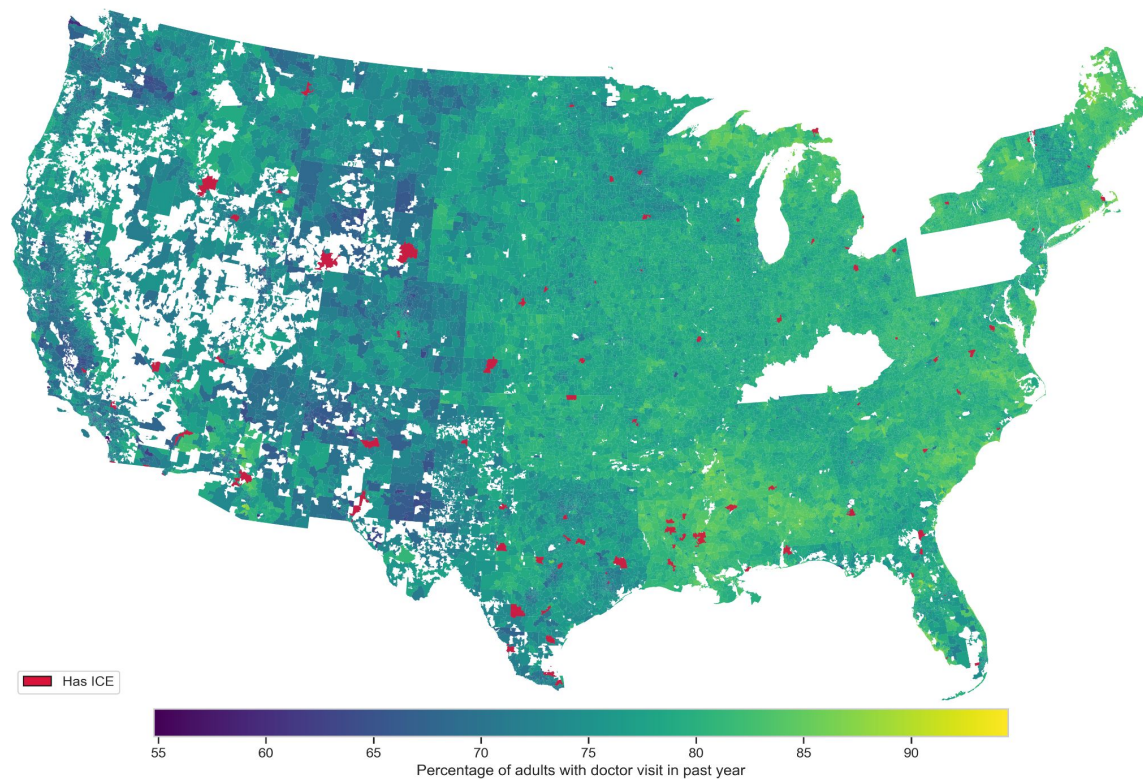
Figure 1: Correlation matrix. Note: Correlations below 0.01 are suppressed.

- **Hispanic/latino folks and non-citizens are less likely to have had a recent doctor's visit**
- **Proximity to detention facility slightly associated with reduced health care utilization**
- **Size of the population in ICE detention facilities not strongly correlated with doctor visits**



Doctor visits vs. ICE facilities

Figure 2: Snapshot of rates of adults who reported a doctor visit within the past year and ICE detention centers, 2023.
Note: Some ZCTAs are missing from the data and are left blank.



Results

Model Results

Model (n = 185,501)	Coefficient	Std. Error	P-value	R-squared (Within)	R-squared (Overall)
Model 1: Binary (TWFE)	-0.177	-0.113	0.1191	-0.0082	N/A
Model 2: Log Distance	-0.572***	-0.026	0	N/A	0.2792
Model 3: Regional Presence (<=50m)	0.785***	-0.039	0	N/A	0.2741
Model 4: 0-10 Miles	0.987***	-0.079	0	N/A	0.2741
Model 4: 10-25 Miles	0.874***	-0.052	0	N/A	0.2741
Model 4: 25-50 Miles	0.682***	-0.044	0	N/A	0.2741

Results

No measurable healthcare effect due to ICE Detention Facility presence over time

- When controlling for national shocks and static geographic differences, the localized increase in ICE presence and activity within a specific ZCTA did not independently drive detectable decreases in local healthcare utilization.

The Urban Healthcare Paradox

- Paradoxically, the models indicated that living closer to an ICE facility was associated with higher levels of utilization. Rather than disproving the enforcement related fear, this highlights the confounder that ICE facility presence is acting as a proxy for robust urban healthcare access.

The Critical Need for Fixed Effects

- The difference between the spatial (false positive correlation) and the TWFE (no significant effect) model shows that most variance in healthcare access is driven by inherent geographic advantages rather than shifting demographics, demonstrating exactly why the Two-Way Fixed Effects model was necessary to strip away the "urban advantage" and isolate true temporal impacts.

Limitations

Limitations

Response

01 Spatial models didn't take healthcare access into consideration.

02 Missing geographies in CDC PLACES data

03 Sparse data on detention facilities



Future research should incorporate additional controls for healthcare access, improve geographic coverage, and bring in more balanced data on local immigration enforcement—not just detention centers.

Conclusions

No statistically significant evidence that ICE detention facility presence reduces aggregate routine medical visits.

Methodological Findings

Spatial models alone are insufficient due to the "urbanicity confounder". Enforcement centers are often located in hubs with high healthcare access.

Two-Way Fixed Effects are critical to isolate enforcement impact.

Aggregation Limits

Aggregate statistical stability does not disprove individual fear/avoidance.

Vulnerable populations are "mathematically eclipsed" by the broader citizen population in ZCTA-level data.

Policy Implications

Officials must prioritize targeted community outreach over high-level geographic assumptions.

Proximity to a hospital does not guarantee the perceived safety to seek care.

Q&A

References

1. Transactional Records Access Clearinghouse. 2026. Detention facilities average daily population.
2. Centers for Disease Control and Prevention. 2025. PLACES: Local data for better health, ZCTA data, 2025 release.
3. U.S. Census Bureau. 2018-2023. American community survey 5-year estimates: Comparison profiles 5-year.

For access to the data and code used in this analysis, visit our GitHub repository:
[data698-master-research-project](#).